

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Modular Unit 1

Adaptive Classified

OL-T14 · Exchange Rates and Best Buys

With Answers

Questions with full worked solutions

■ 3 pages

FAMILY MAP

Exchange Rates Best Buys

F01

Compare best value across currencies or pack sizes

1 Q | 4 marks

- 12 Astrid wants to buy some oil.
She can buy the oil from either Dane Oil or Arctic Oil.

Here is information about the price that each company will charge Astrid.

Dane Oil	Arctic Oil
(4.2×10^5) litres for 2 500 000 Krone	(8.6×10^5) litres for 770 000 Dollars

Astrid wants to get the better value for money for the oil.

$$1 \text{ Dollar} = 6.57 \text{ Krone}$$

From which company should she buy her oil, Dane Oil or Arctic Oil?
You must show your working.

Worked solution

January 2020 | Question 12 | 4 marks

Family: Compare best value across currencies or pack sizes • Variant: Currency-normalised best buy

Method / working

1. Dane cost/litre = $2,500,000 / (4.2 \times 10^5) = 5.952\dots$ Krone.
2. Convert Arctic cost: $770,000 \times 6.57 = 5,058,900$ Krone.
3. Arctic cost/litre = $5,058,900 / (8.6 \times 10^5) = 5.882\dots$ Krone.
4. Arctic Oil is cheaper per litre.

Final answer: Arctic Oil